

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA0713

COURSE NAME: COMPUTER NETWORKS

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Y. Guru Charan/192565020 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Y. Guru Charan

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial X^3+1 .

i) Use polynomial long division to determine the message that should be transmitted.

ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?

Answer: CRC Error Detection

Given Message: 11100011

Generator Polynomial: $x^3 + 1$ (Binary: 1001)

(i) CRC Calculation using Polynomial Division

Step 1: Append 3 zeros (degree of polynomial)

11100011 $\rightarrow$ 11100011000

Step 2: Perform modulo-2 division using XOR.

Final Remainder = 011

Step 3: Transmitted Message = Original Message + CRC
 $= 11100011 + 011 = 11100011011$

(ii) Error Detection at Receiver

When the transmitted message travels through the communication channel, noise may introduce errors. In this case, the **leftmost bit is inverted**.

- **Transmitted message:**
11100011011
- **Received message (after error):**
01100011011

Step 1: Receiver Performs CRC Division

The receiver uses the same generator polynomial (**1001**) and divides the received message

- Perform **modulo-2 division (XOR operation)**

Step 2: Check the Remainder

- If the received data is error-free $\rightarrow$ remainder = **000**
- If any error occurs $\rightarrow$ remainder $\neq$ **000**

~ In this case, after division:

Remainder is non-zero

Step 3: How Receiver Detects Error

- CRC works based on the principle that a valid codeword is **exactly divisible** by the generator polynomial
- Due to bit inversion, the received message is **no longer divisible** by 1001
- This produces a **non-zero remainder**, indicating corruption

Step 4: Final Interpretation

- The receiver **does not correct the error**, only detects it
- Once an error is detected:
 - The receiver may **discard the frame**
 - Or request **retransmission (ARQ protocol)**

Conclusion

Cyclic Redundancy Check (CRC) is a powerful and widely used **error detection technique** in data communication systems. It ensures data integrity by appending a set of redundant bits (CRC bits) to the original message before transmission.

2.

a) A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

b) A network has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal- to-noise ratio is 3162. Find the channel capacity

Answer:

2 (a) Throughput Calculation

Given:

- Bandwidth = 10 Mbps
- Frames per minute = 12,000 frames
- Bits per frame = 10,000 bits

Step 1: Find total bits transmitted per minute

$$\text{Total bits} = 12000 \times 10000 = 120,000,000 \text{ bits/minute}$$

Step 2: Convert to bits per second (Throughput)

$$\text{Throughput} = \frac{120,000,000}{60} = 2,000,000 \text{ bits/sec}$$

Throughput = 2 Mbps

Explanation

Even though the network can handle up to 10 Mbps, the actual data being transmitted is only 2 Mbps. This happens because:

- Not all bandwidth is used efficiently
- There may be delays, overhead, or idle time

So, throughput is the real performance, not the maximum capacity

2 (b) Channel Capacity (Shannon Formula)

Given:

- Bandwidth (B) = 3000 Hz
- Signal-to-noise ratio (SNR) = 3162

Step 1: Use Shannon Capacity Formula

$$C = B \log_2(1 + SNR)$$

Step 2: Substitute values

$$C = 3000 \log_2(1 + 3162)$$

$$C = 3000 \log_2(3163)$$

Step 3: Convert $\log_2$

$$\log_2(3163) \approx 11.62$$

Step 4: Final Calculation

$$C = 3000 \times 11.62 = 34,860 \text{ bps}$$

Channel Capacity $\approx 34.86 \text{ kbps}$

Explanation

This means the channel can transmit up to $\sim 34.86 \text{ kbps}$ without errors under ideal conditions.

Even if we try to send data faster:

- Noise in the channel will cause errors
- Data may get corrupted

So, this value represents the maximum reliable speed of communication.

Overall Conclusion

- Throughput (2 Mbps) shows actual data transfer rate
- Channel capacity (34.86 kbps) shows theoretical maximum limit based on noise
- In real networks:
- Throughput is usually less than bandwidth

3. A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the Hamming (12,8) error-correcting code.

Answer:

Hamming (12,8) Error-Correcting Code in Hospital Monitoring System

Introduction

In a multispecialty hospital, patient-monitoring devices continuously send vital data like heart rate and blood pressure to a central system. Since the transmission happens over a wireless channel, it is affected by electromagnetic interference, which may flip a bit during transmission.

To avoid incorrect medical decisions and without waiting for retransmission, the hospital uses Hamming (12,8) error-correcting code.

What is Hamming (12,8) Code?

- It is an error detection and correction technique
- Converts 8-bit data into 12-bit codeword
- Adds 4 parity bits to detect and correct single-bit errors

Structure of Hamming (12,8)

- Total bits = 12
- Data bits = 8
- Parity bits = 4

👉 Parity bits are placed at positions:

1, 2, 4, 8 (powers of 2)

👉 Data bits are placed in remaining positions:

3, 5, 6, 7, 9, 10, 11, 12

Working Process

Step 1: Sender Side (Encoding)

- Take 8-bit patient data
- Place data bits in correct positions
- Calculate parity bits using even parity
- Form a 12-bit codeword

Step 2: Transmission

- Codeword is sent through wireless channel
- Due to interference, one bit may change

Step 3: Receiver Side (Error Detection & Correction)

- Receiver recalculates parity bits
- Generates a syndrome value
- Syndrome indicates the position of error

If syndrome = 0 → No error

If syndrome $\neq$ 0 → Error position is identified

Step 4: Error Correction

- Flip the bit at error position
- Recover the original correct data

Example (Simple Understanding)

- Sent codeword: 101100101011
- Received codeword: 101100001011 (1-bit error)

Receiver calculates syndrome → finds error position

Corrects the bit automatically

Why Hamming Code is Used in Hospital?

- Immediate correction (no delay)
- No need for retransmission
- Ensures accurate patient data
- Critical for real-time monitoring systems

Advantages

- Detects and corrects single-bit errors
- Fast and efficient
- Improves data reliability
- Suitable for wireless communication

Limitations

- Cannot correct multiple-bit errors
- Slight increase in data size (overhead)
- More complex than simple parity

Conclusion

Hamming (12,8) code plays a crucial role in ensuring reliable and accurate data transmission in hospital monitoring systems. By automatically detecting and correcting single-bit errors, it helps doctors receive correct patient information in real time, which is essential for timely medical decisions. This makes it highly suitable for critical healthcare applications where accuracy and speed are important.

4.

a) A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address 172.16.10.95 falls within the same subnet range based on the given configuration.

b) A network uses the Internet Protocol version 6 address

2001:0db8:0000:0000:0000:ff00:0042:8329. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Answer:

(a) Subnetting Calculation

Given:

- IP Address: 172.16.10.0/24
- New Subnet Mask: /27

1. Number of Subnets

- Borrowed bits = $27 - 24 = 3$ bits
- Number of subnets = $2^3 = 8$

👉 Total Subnets = 8

2. Usable Hosts per Subnet

- Host bits = $32 - 27 = 5$ bits
- Total hosts = $2^5 = 32$
- Usable hosts = $32 - 2 = 30$

👉 Usable Hosts per Subnet = 30

3. Subnet Ranges (/27)

Each subnet size = 32 addresses

Subnet	Network Address	Address Range	Broadcast
1	172.16.10.0	1 – 30	31
2	172.16.10.32	33 – 62	63
3	172.16.10.64	65 – 94	95
4	172.16.10.96	97 – 126	127
...	...	...	...

4. IP: 172.16.10.78

- Falls in range 65 – 94

👉 Subnet = 172.16.10.64/27

- Network Address = 172.16.10.64
- Broadcast Address = 172.16.10.95

5. IP: 172.16.10.95

- 95 is the broadcast address of the same subnet

👉 Yes, it belongs to the same subnet
(but cannot be assigned to a host)

Final Answer (a):

- Subnets = 8
- Hosts per subnet = 30
- 172.16.10.78 → Subnet 172.16.10.64/27
- Network = 172.16.10.64
- Broadcast = 172.16.10.95
- 172.16.10.95 → Same subnet (broadcast address)

4 (b) IPv6 Addressing

Given Address:

2001:0db8:0000:0000:0000: ff00:0042:8329

1. Compressed Form

Rules:

- Remove leading zeros
- Replace consecutive 0000 with:

👉 Compressed Address:

2001:db8::ff00:42:8329

2. Expanded Form

👉 Expanded Address:

2001:0db8:0000:0000:0000: ff00:0042:8329

3. Network Prefix (/64)

- First 64 bits = Network

👉 Network Prefix:

2001:db8:0:0: :/64

(or)

2001:db8: :/64

4. Number of Hosts

- Host bits = $128 - 64 = 64$ bits
- Total hosts = 2^{64}

👉 Total Hosts = 18,446,744,073,709,551,616

Final Answer (b):

- Compressed = 2001:db8: ff00:42:8329
- Expanded = 2001:0db8:0000:0000:0000: ff00:0042:8329
- Network Prefix = 2001:db8: :/64
- Total Hosts = 2^{64} ($\approx 1.84 \times 10^{19}$)

5. A university is conducting an online semester examination for 8,000 students simultaneously. At exactly 10:00 AM, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a 10 Gbps high-speed network, while students access the examination portal using different internet connections such as fiber broadband (100 Mbps), home Wi-Fi (20 Mbps), and mobile networks (5–10 Mbps). During the first few minutes of the examination, many students with slower internet connections report incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and process packets at the same speed. This results in receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers. Analyze the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission

Answer:

Analysis of the Networking Problem

Primary Problem:

👉 The main issue is Flow Control Problem (Receiver Overload)

Explanation:

In this scenario:

- The server (sender) transmits data at 10 Gbps (very high speed)
- Students (receivers) have much slower connections (5–100 Mbps)

👉 This creates a speed mismatch between sender and receivers

What Happens Due to This Mismatch:

1. Receiver Buffer Overflow

- Student devices cannot process incoming packets quickly
- Their buffer becomes full → packets get dropped

2. Packet Loss & Retransmissions

- Lost packets must be retransmitted
- Increases network traffic unnecessarily

3. Delayed Acknowledgments (ACKs)

- Slow receivers send ACKs late
- Sender continues sending data aggressively

4. Congestion Build-Up

- Multiple retransmissions increase traffic load
- Leads to network congestion

5. Poor User Experience

- Incomplete downloads
- Freezing of portal
- Delays in receiving question papers

Root Cause:

👉 Lack of proper flow control mechanism

- The sender is not adjusting its transmission rate based on receiver capacity
- No effective use of TCP flow control (sliding window / receive window)

Concept Involved:

👉 Flow Control (End-to-End Control)

Ensures:

- Sender does not overwhelm the receiver
- Data is sent at a rate the receiver can handle

Solution / Prevention:

1. Use TCP Flow Control (Sliding Window)

- Adjust sending rate based on receiver buffer size

2. Congestion Control Mechanisms

- Slow Start
- Congestion Avoidance

3. Rate Limiting / Throttling

- Server should limit transmission speed

4. Content Distribution Techniques

- Use CDN or multiple servers to distribute load

5. Buffer Management

- Improve receiver-side buffering

Final Conclusion:

👉 The problem occurs due to mismatch between high sender transmission rate and low receiver processing speed, leading to flow control failure, causing buffer overflow, retransmissions, and network congestion.